Final Data Analysis Project Analysis of Longitudinal Data

A Longitudinal Analysis and Comparative Study with Boeing's Stock Price

1. Introduction

Boeing's stock performance reflects both internal operations and external market dynamics, making it a critical benchmark for trends in the aerospace industry. This study examines Boeing's stock behavior over a five-year period (2019–2024), focusing on key metrics such as Daily Return, Cumulative Return, and Volatility. The analysis evaluates how significant political, financial, and company-specific events, such as the 737 MAX grounding and the COVID-19 pandemic, have influenced stock trends and variability over time. By employing a longitudinal approach, this study aims to uncover the temporal impacts of these events, providing insights into Boeing's unique market behavior.

2. Data and Methodology

This study evaluates how significant events influence Boeing's stock price over time using a longitudinal approach. Daily data from Google Finance (2019–2024) forms the basis of the analysis, with key metrics including Daily Return, Cumulative Return, Volatility, Moving Averages, and RSI_14. Binary indicators for events such as the COVID-19 pandemic and 737 MAX grounding assess their temporal impacts, while interaction terms evaluate how events modify relationships between predictors and outcomes. A fixed-effects regression model controls for time-invariant factors, isolating the effects of predictors and events. Models are incrementally fitted and evaluated using adjusted R^2, AIC/BIC, and residual analysis. **Table 1** and **Figure 1** illustrate Boeing's metrics and heightened volatility during disruptions.

3. Results and Analysis

To evaluate how significant events influence Boeing's stock performance, eight models were fitted and compared. Key performance metrics, including AIC and Adjusted R^2, are summarized in **Table 2**. In the table, however, Lasso and Fixed-Effects Models are excluded as their AIC values are not directly comparable. Among these six models, the Volatility Model demonstrated the best fit, achieving the lowest AIC (4696.737). The Interaction Model and Full Model followed closely, with AIC values of 7029.458 and 7030.891, respectively. Adjusted R^2 values ranged from 0.202 (Baseline Model) to 0.223 (Volatility Model), indicating that event indicators and interaction terms enhance explanatory power.

The Volatility Model focuses on stock variability, achieving the lowest AIC among all models. This model best captures Boeing's stock dynamics. The Volatility Model effectively highlights stock fluctuations during crises, such as the 2020 market crash, reflecting increased investor uncertainty and market volatility. This suggests that Boeing's stock returns were strongly influenced by periods of elevated risk and disruption.

Residual analysis for the Fixed-Effects Model (**Figure 2**) further supports the model's validity by showing random noise without patterns, indicating that the model effectively captures the systematic variation in the stock. Together, these findings emphasize how both systemic events (e.g., COVID-19, the 737 MAX grounding) and company-specific incidents significantly impact Boeing's stock performance, driving not only changes in returns but also increases in volatility during periods of market turmoil.

The Fixed-Effects Model, designed to control for unobserved, time-invariant factors, achieved a within- R^2 of 0.215. While its AIC could not be computed for direct comparison, this model effectively isolates the impact of time-varying predictors and event indicators, making it a critical tool for understanding Boeing's longitudinal stock dynamics.

Events such as the COVID-19 pandemic, the 737 MAX grounding, and the 2020 market crash were associated with substantial changes in stock performance. **Figure 3** illustrates these periods, with shaded regions marking key disruptions. The green shaded area highlights the adverse effects of the 737 MAX grounding, while the red region shows volatility during COVID-19. Coefficient estimates from the Fixed-Effects Model (**Table 3**) quantify these impacts. Event_COVID, with a positive coefficient (1.894), reflects temporary rebounds during recovery periods despite initial disruptions. Conversely, Event_737_MAX (-0.691) and Event_Market_Crash (-2.773) capture the prolonged negative impacts of these crises. These findings underline the pivotal role of systemic disruptions and company-specific incidents in shaping Boeing’s stock trajectory.

The Event Model showed significant improvement over the Baseline Model too, with a reduction in AIC by over 50 points (**Table 2**), further highlighting the importance of including event indicators in explaining stock performance.

4. Discussion and Conclusion

Significant events like COVID-19, the 737 MAX grounding, and the 2020 market crash significantly impacted Boeing’s stock performance. The Volatility Model best captured these effects, highlighting the role of volatility. The Fixed-Effects Model provided insights into time-varying impacts. Overall, Boeing’s stock is strongly influenced by external and company-specific events, with volatility playing a key role.

Appendix

Figure 1

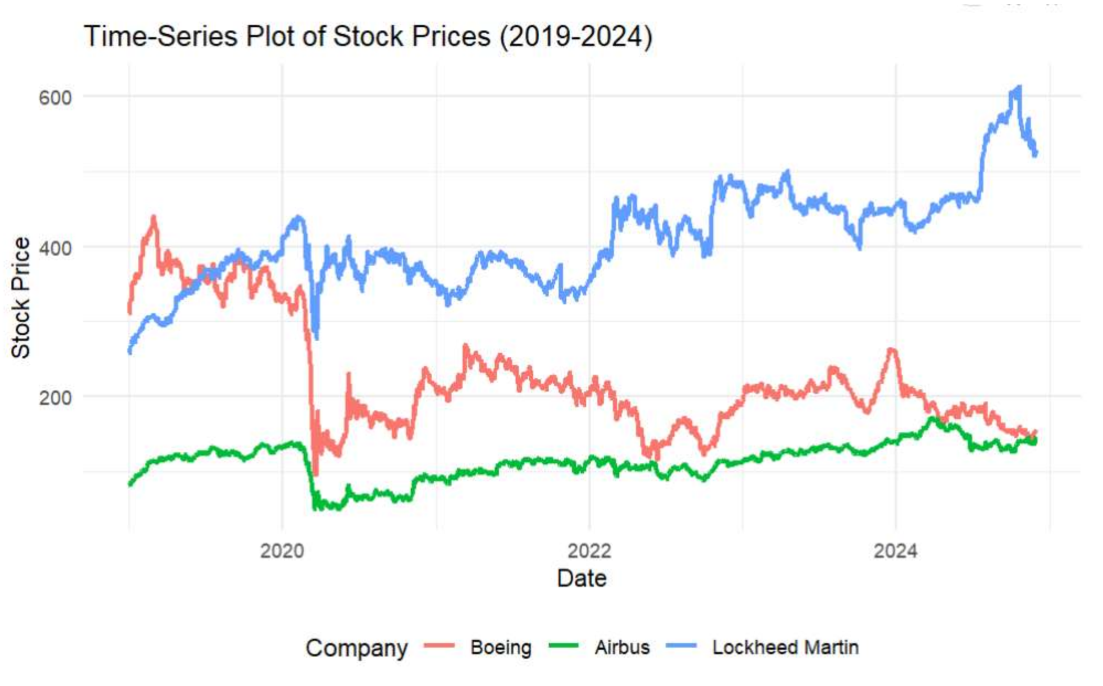


Figure 2

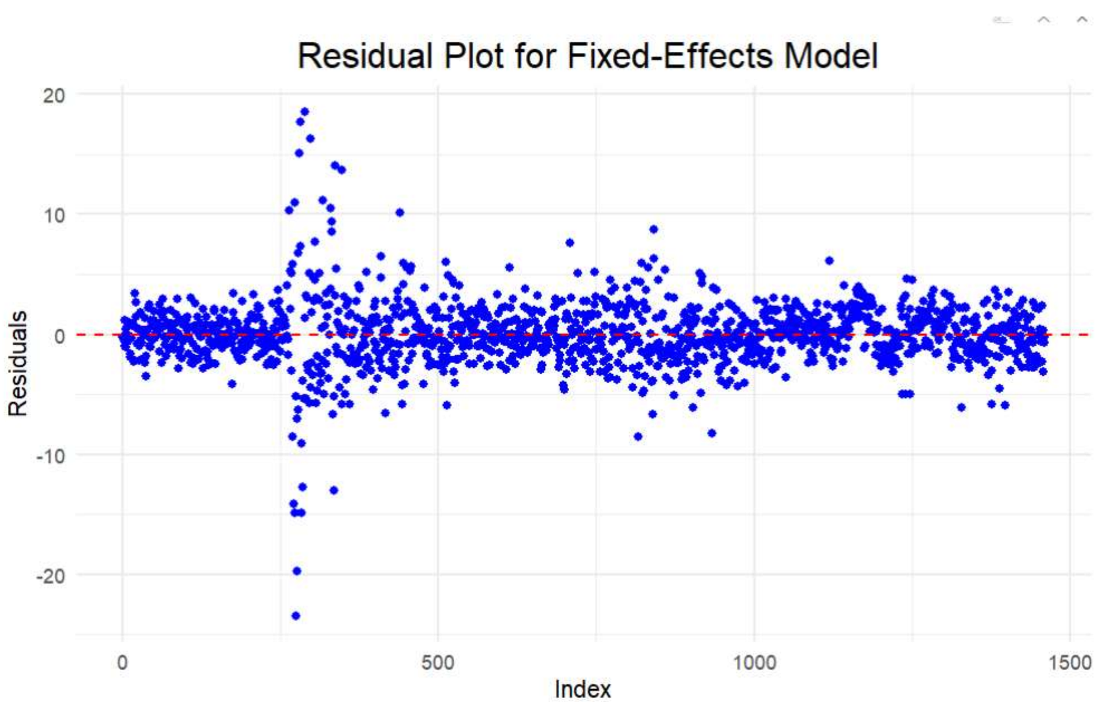
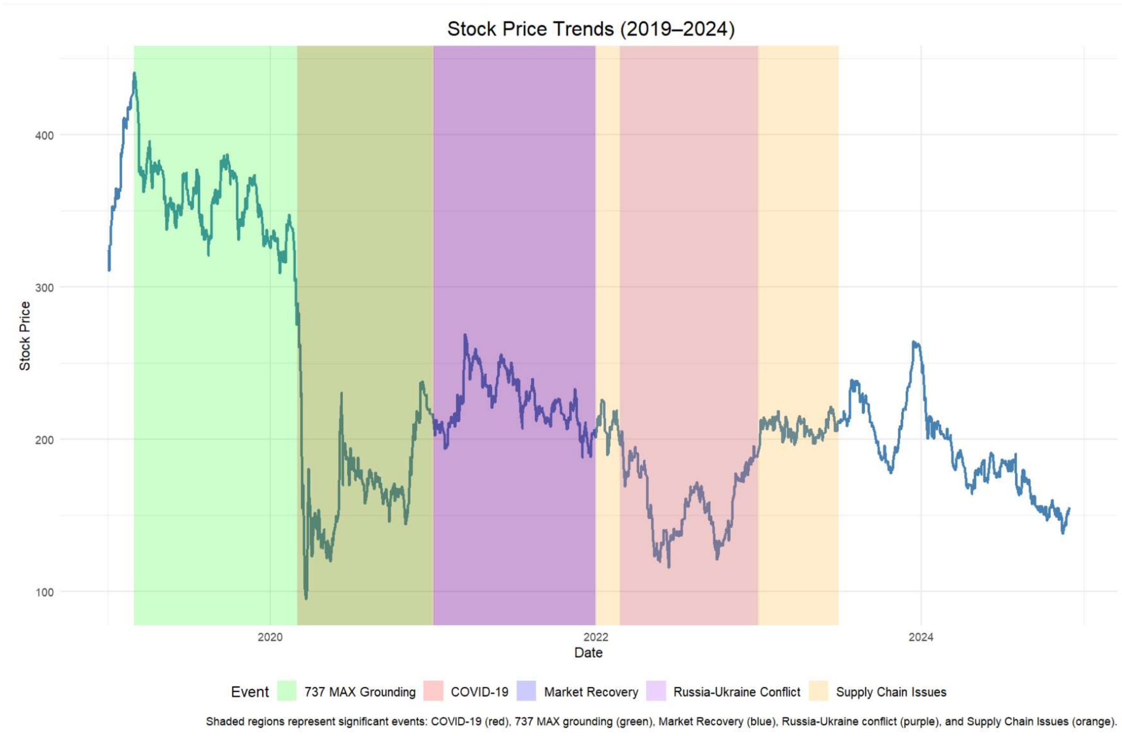


Figure 3



Shaded regions represent significant events: COVID-19 (red), 737 MAX grounding (green), Market Recovery (blue), Russia-Ukraine conflict (purple), and Supply Chain Issues (orange).

Table 1

Date	Close	Daily_Return	MA_7	MA_30	Volatility_7	Volatility_30	Cumulative_Return	RSI_14	EMA_7	EMA_30	Event_737_MAX	Event_COVID	Event_Market_Crash	Event_Trade_War	Event_Defense_Budget	Event_EU_Regulations	Event_Russia_Ukraine	Event_WTO_Dispute	Event_Earnings	Event_Bailouts	Event_787_Delays	Event_Technical_Incidents	Event_A380_Halt
2019-01-02	323.81	0.000000000	NA	NA	NA	NA	0.000000000	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-03	310.90	-3.986905902	NA	NA	NA	NA	-0.039869059	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-04	327.08	5.204245738	NA	NA	NA	NA	0.010098515	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-07	328.11	0.314907668	NA	NA	NA	NA	0.013279392	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-08	340.53	3.785315900	NA	NA	NA	NA	0.051635218	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-09	343.83	0.969077614	NA	NA	NA	NA	0.061826380	NA	NA	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-10	352.61	2.553587529	332.4100	NA	NA	NA	0.088941046	NA	332.4100	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-11	352.90	0.082243839	336.5657	NA	2.9861103	NA	0.089836633	NA	337.5325	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-14	350.36	-0.719750638	342.2029	NA	2.1706361	NA	0.081992526	NA	340.7394	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-15	352.24	0.536590935	345.7971	NA	1.5608529	NA	0.087798400	NA	343.6145	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-16	352.06	-0.051101522	349.2186	NA	1.5962707	NA	0.087242519	NA	345.7259	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-17	359.09	1.996818724	351.8700	NA	1.1652525	NA	0.108952781	NA	349.0669	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-18	364.73	1.570636888	354.8557	NA	1.2041903	NA	0.126370402	NA	352.9827	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-22	357.90	-1.872618101	355.6114	NA	1.3188498	NA	0.105277786	NA	354.2120	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-23	358.61	0.198379436	356.4271	NA	1.3175519	NA	0.107470430	71.82639	355.3115	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-24	358.27	-0.094810518	357.5571	NA	1.2619165	NA	0.106420432	71.49800	356.0511	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-01-25	364.20	1.655176264	359.2657	NA	1.3600004	NA	0.124733640	73.75204	358.0884	NA	0	0	0	1	0	0	0	1	1	0	0	0	
2019-01-28	362.97	-0.337726524	360.8243	NA	1.3829894	NA	0.120935116	72.47181	359.3088	NA	0	0	0	1	0	0	0	1	1	0	0	0	
2019-01-29	364.91	0.534479434	361.6557	NA	1.2090416	NA	0.126926284	73.26022	360.7091	NA	0	0	0	1	0	0	0	1	1	0	0	0	
2019-01-30	387.72	6.250856376	364.9400	NA	2.5831730	NA	0.197368827	80.37662	367.4618	NA	0	0	0	1	0	0	0	1	1	0	0	0	
2019-01-31	385.62	-0.541627979	368.9000	NA	2.3861290	NA	0.190883543	78.31029	372.0014	NA	0	0	0	1	0	0	0	1	1	0	0	0	
2019-02-01	387.43	0.469373995	373.0171	NA	2.3713104	NA	0.196473240	78.81580	375.8585	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-04	397.00	2.470123635	378.5500	NA	2.3478983	NA	0.226027609	81.29770	381.1439	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-05	410.18	3.319899244	385.1186	NA	2.4483839	NA	0.266730490	84.06640	388.4029	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-06	411.11	0.226729728	391.9957	NA	2.3768811	NA	0.269602545	84.24365	394.0797	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-07	405.17	-1.444868770	397.7471	NA	2.6563995	NA	0.251258454	78.25570	396.8523	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-08	404.91	-0.064170595	400.2029	NA	1.6817458	NA	0.250455514	77.99437	398.8667	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-11	403.95	-0.237089724	402.8214	NA	1.6499132	NA	0.247490813	76.97231	400.1375	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-12	410.72	1.675949994	406.1486	NA	1.6871947	NA	0.268398135	79.05662	402.7831	NA	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-13	410.58	-0.034086482	408.0886	367.9833	1.5458679	NA	0.267965782	78.89759	404.7324	367.9833	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-14	409.82	-0.185103999	408.0371	370.8503	0.9167928	2.042579	0.265618727	77.98048	406.0043	370.6825	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-15	417.97	1.988677956	409.0171	374.4193	1.1926530	1.840188	0.290787808	80.58655	408.9957	373.7333	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-19	416.26	-0.409120272	410.6014	377.3920	0.9965051	1.676866	0.285506933	78.48760	410.8118	376.4769	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-20	421.55	1.270840340	412.9786	380.5067	1.0223508	1.676015	0.301843674	80.20525	413.4963	379.3849	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-21	417.58	-0.941762543	414.9257	383.0750	1.1438547	1.611713	0.289583398	75.34331	414.5172	381.8491	0	0	0	1	0	0	0	1	0	0	0	0	
2019-02-22	424.05	1.549403707	416.8300	385.7490	1.1226202	1.618590	0.309564251	77.71431	416.9004	384.5717	0	0	0	1	0	0	0	1	0	0	0	0	

Table 2

Table 2: Model Performance Summary

Model	AIC	Adj_R2
Baseline	7144.611	0.2023457
Event	7094.262	0.2099996
Interaction	7029.458	0.2150000
Time Trend	7086.096	0.2110000
Volatility	4696.737	0.2230000
Full	7030.891	0.2180000

Table 3

Table 3: Fixed-Effects Model Coefficients

	s1
(Intercept)	-3.3601119
RSI_14	0.0188938
MA_7	-0.0228894
MA_30	0.0322188
Time	0.0000314
Event_COVID	1.8941395
Event_737_MAX	-0.6919120
Event_Market_Crash	-2.7728808
Event_Trade_War	-0.7006387
Event_Defense_Budget	0.5589249
Event_EU_Regulations	-2.1532992